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Test Booklet Series

A

T. B. C. : GEO – 2/21

TEST BOOKLET

GEOLOGY

PAPER – II

Sl. No. 20365

Time Allowed : 3 Hours

Maximum Marks : 300

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5. This Test Booklet contains **150** items (questions). Each item (question) comprises four responses (answers). You have to select the correct response (answer) which you want to mark (darken) on the Answer Sheet. In case, you feel that there is more than one correct response (answer), you should mark (darken) the response (answer) which you consider the best. In any case, choose **ONLY ONE** response (answer) for each item (question).
6. You have to mark (darken) all your responses (answers) **ONLY** on the **separate Answer Sheet** provided, by **using BALL POINT PEN (BLUE OR BLACK)**. See instructions in the Answer Sheet.
7. All items (questions) carry equal marks. All items (questions) are compulsory. Your total marks will depend only on the number of correct responses (answers) marked by you in the Answer Sheet. **There shall be negative marking for wrong answers with a deduction of 0.5 marks for every wrong answer.**
8. Before you proceed to mark (darken) in the Answer Sheet the responses (answers) to various items (questions) in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per the instructions sent to you with your **Admission Certificate**.
9. After you have completed filling in all your responses (answers) on the Answer Sheet and after conclusion of the examination, you should hand over to the Invigilator the *Answer Sheet* issued to you. You are allowed to take with you the candidate's copy / second page of the Answer Sheet along with the **Test Booklet**, after completion of the examination, for your reference.
10. Sheets for rough work are appended in the Test Booklet at the end.

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SEAL

1. The generic term describring those downward movements of slope-forming material as a result of shear failure occurring along a well-defined shear plane is :

(A) Mass movement

(B) Landslide

(C) Creep

(D) Avalanche

2. Sinuosity Index of a meandering river is greater than the threshold value of :

(A) 1.4

(B) 1.5

(C) 1.6

(D) 1.2

3. Aeolian activity is primarily controlled by :

(i) Transport capacity

(ii) Sediment supply

(iii) Hot desertic conditions

(iv) Sediment availability

Choose the correct set of answer from the following options :

(A) (i), (ii), (iii)

(B) (i), (ii), (iv)

(C) (i), (iii), (iv)

(D) (i), (ii), (iii), (iv)

4. Which of the following indices belong to 'Tectonic Activity Indices' used for morphotectonic analysis of a drainage basin ?

(i) Drainage basin asymmetry

(ii) Stream Length-Gradient Index (SL Index)

(iii) Mountain-front sinuosity

(iv) Ratio of valley floor width to valley height

Choose the correct set of answer from the following options :

(A) (i), (ii) and (iv)

(B) (ii) and (iv)

(C) (i), (ii), (iii) and (iv)

(D) (ii), (iii) and (iv)

5. An isolated erosional rock mass comprising a slender stem, neck or column supporting a wider cap is known as :

(i) Pedestal rocks

(ii) Mushroom rocks

(iii) Balanced rocks

(iv) Perched blocks

Choose the correct set of answer from the following options :

(A) (i)

(B) (i) and (ii)

(C) (i), (ii) and (iii)

(D) (i), (ii), (iii) and (iv)

6. Optically Stimulated Luminescence (OSL) dating methods are useful for determing the age of :

(i) Naturally occurring sediments

(ii) Man-made materials like pottery

(iii) Intrusive igneous rocks

(iv) Metasedimentary rocks

Choose the correct set of answer from the following options :

(A) (i), (ii) and (iii)

(B) (i) and (ii)

(C) (iii) and (iv)

(D) (i) and (iii)

7. Fluvial megafans :

(i) constitute volumetrically significant depositional elements of sedimentary basins adjacent to mountain belts

(ii) are related to the emergence of a large mountain river onto a low-relief alluvial plain

(iii) are also known as alluvial fans

(iv) are distinguished by their greater size and lower slope

Choose the correct set of answer from the following options :

(A) (i), (ii), (iii)

(B) (ii), (iii), (iv)

(C) (i), (ii), (iv)

(D) (i), (iii), (iv)

8. If a network of parallel streams is developed along strike and dip of the formations, the drainage pattern is referred to as :

(A) Parallel

(B) Rectangular

(C) Trellis

(D) Dendritic

9. The lithified portions of alluvial fans are termed as :

(A) Bajada

(B) Agglomerate

(C) Fanglomerate

(D) Orthoconglomerate

10. A long, narrow ridge or a series of hills with a narrow crest and steep slopes of nearly equal inclination on both flanks created by the differential erosion of outcropping sedimentary strata is termed as :

(A) Cuesta

(B) Hogback

(C) Mesa

(D) Butte

11. What is responsible for the blue colour of the sky ?
    (A) Mie Scattering
    (B) Non-Selective Scattering
    (C) Rayleigh Scattering
    (D) Water droplets present in the sky

12. The vibrational energy of an object is associated with the _____ of the electromagnetic spectrum.
    (A) UV and Photo UV region
    (B) SWIR and TIR region
    (C) Visible and NIR region
    (D) Microwave and Radio wave region

13. In Sun Synchronous orbit, the satellite is synchronized with :
    (A) Particular area of the Earth
    (B) Local Solar time
    (C) Stationary on the Equator
    (D) Stationary on the Pole

14. Clarity of ground object will depend upon :
    (A) Spatial Resolution
    (B) Spectral Resolution
    (C) Temporal Resolution
    (D) Radiometric Resolution

15. Fiducial marks recorded on Aerial photographs are used to determine the :
    (A) Flight path
    (B) Scale of Aerial photographs
    (C) Principal points
    (D) Flight height

16. Major difference between Real Aperture Radar and Synthetic Aperture Radar is in their :
    (A) Size of the Antenna
    (B) Source of energy
    (C) Recording of data
    (D) None of these

17. The moisture present in the ground surface is shown as _____ tone in the NIR region of imagery.
    (A) white
    (B) grey
    (C) red
    (D) blue

18. In the visible region (green band), leaf reflectance is governed by :
    (A) Leaf moisture
    (B) Leaf pigment
    (C) Leaf cell structure
    (D) None of these

19. In IRS 1D satellite, the spatial resolution of MSS sensor is :
    (A) 1 m
    (B) 30 m
    (C) 79 m
    (D) 23.5 m

20. Name the satellite providing three-dimensional topographic data :

(A) LANDSAT

(B) SRTM

(C) IKONOS

(D) ERS

21. Thrust faults are reverse faults that dip _____.

(A) 45°

(B) $< 45^{o}$

(C) $> 30^{o}$

(D) Both (A) and (B) are correct

22. The vertical angle between the horizontal plane and the axis or line of maximum elongation of a feature is called _____.

(A) dip angle

(B) strike

(C) right-lateral strike slip

(D) plunge

23. In elastic deformation, when strain is proportional to stress is known as :

(A) Smith's law

(B) Poisson's ratio

(C) Hooke's law

(D) None of these

24. The line of maximum curvature in a fold is known as _____.

(A) trough

(B) axis

(C) crest

(D) hinge

25. A fold in which the anticlines become sharper and synclines become broader with depth is known as :

(A) Parallel fold

(B) Similar fold

(C) Recumbent fold

(D) Box fold

26. A valley formed by the downward displacement of a fault bounded block is known as :

(A) Nappe

(B) Fault escarpment

(C) Graben

(D) All of these

27. The vertical angle between the horizontal plane and the axis or line of maximum elongation of a feature is called _____.

(A) dip angle

(B) strike

(C) right-lateral strike slip

(D) plunge

28. A fault in which the movement is parallel to the dip of the fault is called :

(A) Step fault

(B) Dip-slip fault

(C) Strike-slip fault

(D) None of these

29. A type of unconformity in which the beds above and below are parallel is termed as :

(A) Parallel unconformity

(B) Discontinuity

(C) Non-conformity

(D) Disconformity

30. The distance between two points on a 1 : 25,000 map is 1.5 cm than the true distance between these points on the field will be :

(A) 375 m

(B) 3750 km

(C) 3750 cm

(D) 37500 mm

31. A sulphide vein dips 50° toward N45°W. Its apparent dip toward of N20°W will be around _____.

(A) 65°

(B) 60°

(C) 0°

(D) 45°

32. Rocks are deformed in the Himalayan mountain belt commonly forms which deformational structures ?

(A) Normal and reverse faults

(B) Folds and reverse faults

(C) Recumbent folds and normal faults

(D) All of these

33. In a recumbent fold, the axial plane dips about _____.

(A) 45°

(B) 90°

(C) 60°

(D) 0°

34. In the upper crust where The P and T condition is relatively low, the deformational process dominates is _____.

(A) Brittle process

(B) Ductile-brittle process

(C) Plastic deformation

(D) Both (A) and (B) are true

35. Most common angle of normal faults is 45° to vertical.

(A) False

(B) True

(C) Can be true or false

(D) None of these

36. Which river is not flowing through a Rift valley ?

(A) Mahi

(B) Narmada

(C) Krishna

(D) Tungabhadra

37. Find the incorrect pair :

(A) Kermadec Trench – Pacific Ocean

(B) Reykjanes Ridge – Atlantic Ocean

(C) Phoenix Ridge – Arctic Ocean

(D) Carlsberg Ridge – Indian Ocean

38. What will be the location where an earthquake does the maximum damage ?

(A) Epicentre

(B) Focus

(C) 100 km from the epicenter

(D) Isocentre

39. P and S waves are collectively called :

(A) Love waves

(B) Surface waves

(C) Body waves

(D) Rayleigh waves

40. What is the distinguishing feature of the mass above and below a landslide ?

(A) Both masses are stable

(B) Mass above is stable, whereas mass below is unstable

(C) Mass above is unstable, whereas mass below is stable

(D) Both masses are unstable

41. The mountain type which does not belong to the tectonic mountain categories are :

(A) Complex mountains

(B) Volcanic mountains

(C) Fault block mountains

(D) Residual mountains

42. Which of the following is incorrect about Asthenosphere ?

(A) It is present in the upper mantle

(B) It is in solid-state

(C) It is in a semi-molten stage

(D) It is the source of volcanic activity

43. SIMA stands for :

(A) Silicon and Manganese

(B) Strontium and Magnesium

(C) Sulphur and Manganese

(D) Silicon and Magnesium

44. Which statement is true about P wave ?

(A) The P wave reflected from the mantle-core boundary.

(B) The P wave is transmitted through the outer core but reflected from the inner core.

(C) The P wave is transmitted through the outer as well as the inner core.

(D) The P wave is reflected from the crust-mantle boundary.

45. Isostatic equilibrium is :

(A) A buoyancy force pushing the lithosphere up equals the gravitational force pushing it down.

(B) A buoyancy force pushing the lithosphere up is less than the gravitational force pushing it down.

(C) A buoyancy force pushing the lithosphere up is greater than the gravitational force pushing it down.

(D) All the features of Earth's surface are at the same elevation

46. The Upper Cretaceous rocks of the Tiruchchirappalli area has been classified into three Formations viz., Uttatur, Trichinapoli and _____.

(A) Ariyalur Formation

(B) Haimanta Formation

(C) Lilang Formation

(D) Chikkim Formation

47. The Vindhyan Supergroup is divisible into four groups viz., Semri Group, Kaimur Group, Rewa Group and _____.

(A) Bijawar Group

(B) Bhander Group

(C) Gwalior Group

(D) None of these

48. The base of Gondwana sediments in Talcher coalfield, is marked by a pile of _____.

(A) Glacial and Periglacial deposits

(B) Fossiliferous shale

(C) Glossopteris flora

(D) Talcher sandstone

49. Which trace fossils are described from the Lower Devonian Muth Quartzite in Spiti, India ?

(A) **Nautilus**

(B) Unio

(C) Pelecypod

(D) Arthropod

50. Jurassic stratigraphy is primarily based on the use of ammonites as _____.

(A) leaked fossil

(B) index fossil

(C) trace fossil

(D) None of these

51. The Mid-Permian non-coaliferous lithounit sandwitched between two major coal-bearing formations in the Indian Lower Gondwana succession is known as _____ Formation.

(A) Barakar

(B) Lower Kamthi

(C) Barren Measures

(D) Raniganj

52. The Digboi oil field of Assam is on a faulted elongate anticline in rocks of the _____.

(A) Tipam Group
(B) Barail Group
(C) Jaintia Group
(D) All of these

53. As per the concept of 'uniformitarianism', the Earth has always changed in uniform ways and that the '_____.'

(A) past is the key to the present
(B) isostatic equilibrium is a key item
(C) present is the key to the past
(D) None of these

54. The _____ area is popularly known as "Museum of Indian Geology" while referring to the Geology of India.

(A) Spiti
(B) Wiwaliks of kashmir
(C) Palaeozoic sucession of Kashmir
(D) None of these

55. The Vindhyan Supergroup is divisible into four groups viz., Semri Group, Kaimur Group, Rewa Group and _____.

(A) Bijawar group
(B) Bhander group
(C) Gwalior group
(D) None of these

56. _____ is the process of establishing which sedimentary strata are the same age at distant geographical areas by means of their stratigraphic relationship.

(A) Chronostratigraphic correlation
(B) Lithostratigraphic correlation
(C) Biostratigraphic correlation
(D) Stratigraphic correlation

57. Raja Rao of GSI in 1982 classified the stratigraphic sequence of Talcher coalfield, Orissa into Early Permian and Late Permian to Triassic. The Early Permian is again classified into which formations ?

(A) Lower Kamthi
(B) Talchir and Karharbari
(C) Damuda Group
(D) All of these

58. General geological formation of the Aravalli Mountains :

Group/Supergroup	Formation
Aravalli Supergroup[2]	Derbari Group[D]
	Banded Gneissic Complex (BGC)[C]
	Jharol Group[B]
Archean Basement[1]	Delwara Group[A]

Match the correct sequence from older to younger formations. The sequence is :

(A) $1(^{C}), 2(^{A}, ^{D}, ^{B})$
(B) $1(^{C}), 2(^{B}, ^{D}, ^{A})$
(C) $1(^{C}), 2(^{A}, ^{B}, ^{D})$
(D) None of these

59. The Lameta Formation is also known as the _____, is a sedimentary geological formation found in Madhya Pradesh, Gujarat, and Maharashtra, India, associated with the Deccan Traps.

(A) Bagh Beds

(B) Intertrappean Beds

(C) Infratrappean Beds

(D) All of these

60. The boundary in between sedimentary rocks of the Cudappah Supergroup and Archaean rocks found near Tirupati Hills, creating a period of nondepositions and erosion of about 500 Ma has been defined as _____.

(A) The Great Unconformity

(B) Eparchean Unconformity

(C) Paraconformity

(D) None of these

61. It is possible to recognize vertebrate fossils as early as Ordovician on the basis of evidences of brain, paired sensory organs and _____.

(A) anatomy

(B) bodyparts

(C) carbon dating

(D) bone structure

62. The notochord of amphioxus differs from that of vertebrates by extending to the extreme _____ end of the body.

(A) posterior

(B) anterior

(C) open

(D) None of these

63. The only fossil that provides a link beween any of the modern orders and Paleozoic amphibians is _____.

(A) Triadobatrachus

(B) Salamander hynobius

(C) Rhipidistrian fish

(D) None of these

64. The closest known relative of Petrolacosaurus is _____.

(A) Sphenodon

(B) Araeoscelis

(C) Hylonomous

(D) Askeptosaurus

65. Placoderms are a particular assemblage of heavily _____.

(A) Armord jawed fish

(B) Benthonic fish

(C) Plankton

(D) Nekton

66. Foraminifera are _____ organisms.

(A) Autotrophic

(B) both Autotrophic and Heterotrophic

(C) Heterotrophic

(D) None of these

67. Pliopithecus is a genus of _____ found in the Upper Miocene and Lower Pliocene strata of Europe and possibly Asia.

(A) Anthropoids

(B) Bipedal ape-like animals

(C) Lazarus taxa

(D) None of these

68. The earliest mammals are recognized in the _____.

(A) Permian

(B) Upper Triassic and Lower Jurassic

(C) Cambrian to Silurian

(D) Devonian

69. Common kinds of invertebrate fossils include _____, brachiopods, echinoderms, arthropods, and cnidarians.

(A) molluscs

(B) nautiloids

(C) cephalopoda

(D) All of these

70. Ammonite is a characteristic marine fossil of _____ age around the world and considered an index fossil because of its diverse nature.

(A) Palaeozoic

(B) Cambrian to Silurian

(C) Cretaceous-age

(D) None of these

71. The Rugose corals have unique horn-shaped chamber with a wrinkled or rugose wall are called _____.

(A) Scleractinia or Hexacorallia

(B) Tetracoralla or Horn corals

(C) Octocoralla

(D) All of these

72. Brachiopods are considered more important than bivalves in _____ due to their greater global diversity and greater abundance in many fossil assemblages.

(A) Mesozoic ecosystems

(B) Cenozoic ecosystems

(C) Paleozoic ecosystems

(D) Paleocene ecosystems

73. Agglutinated Fossil Foraminifera buried deeply in sedimentary basins can be used to estimate _____, a key factor for hydrocarbon generation.

(A) thermal maturity

(B) diagenesis

(C) burial diagenesis

(D) kerogen content of organic shales

74. Dinoflagellate cysts like Cassiculosphaeridia magma, Hystrichodinium voigti etc. are important in establishing which age of transgressive event in the Krishna-Godavari Basin, India ?

(A) Palaeocene

(B) Jurassic

(C) Early Cretaceous

(D) Late Palaeocene

75. While genous Discocyclina is found in sediments of Danian to Early Eocene but its species Discoclina ranikotensis is only confined to _____.

(A) Late Paleocene

(B) Early Paleocene

(C) Miocene

(D) Late Miocene

76. The composition of a primary magma depends on the following parameters :

(i) Composition of the source material being partially melted

(ii) Relative concentrations of volatiles

(iii) Confining pressure

(iv) Degree of partial melting

Choose the correct set of answer from the following options :

(A) (i), (ii), (iii)

(B) (ii), (iii), (iv)

(C) (i), (iii), (iv)

(D) (i), (ii), (iii), (iv)

77. Match the following in List – I and List – II and choose the correct answer :

	List – I	List – II
(a)	Phaneritic texture	(i) Dolerite
(b)	Ophitic texture	(ii) Gabbro
(c)	Holocrystalline texture	(iii) Spillite
(d)	Pillow structure	(iv) Granite

(A) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(B) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)

(C) (a) – (i), (b) – (ii), (c) – (iv), (d) – (iii)

(D) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)

78. Which of the following is an ultra-alkalic rock ?

(A) Adamellite

(B) Rhyodacite

(C) Transjhemite

(D) Ijolite

79. This texture results from the inter-growth of quartz and plagioclase (usually oligoclase). The inter-growth of quartz is in the form of worm-like rods within the plagioclase. It is found in some granite and metamorphic rocks. It is termed as :

(A) Graphic texture

(B) Myrmekitic texture

(C) Corona texture

(D) Trachytic texture

80. What is the composition of Charnockite ?

(A) Hypersthene + K-feldspar + Quartz

(B) Hornblende + K-feldspar + Quartz

(C) Sillimanite + K-feldspar + Quartz

(D) Epidote + Plagioclase + Quartz

81. An inter-growth texture in which plagioclase contains inclusions of potash feldspar is known as :

(A) Antiperthite

(B) Perthite

(C) Graphic texture

(D) Ophitic

82. Which of the following statements w. r. t. Autometamorphic processes is **not correct** ?

(A) These secondary mineral reactions occur in igneous rocks as they cool, and are not products of a distinct later metamorphic event.

(B) These processes are more common in plutonic rocks than in volcanics.

(C) Diagenetic and weathering processes are also considered to be Autometamorphic.

(D) Most reactions involve minerals at moderate temperatures in an environment in which H_2O is either liberated from residual melt or externally introduced.

83. The ductile shear zone along with the High-grade Great Himalayan Crystalline complex was placed above the Low-grade to unmetamorphosed Lesser Himalayan Sequence is termed as the :

(A) Main Central Thrust (MCT)

(B) Main Frontal Thrust (MFT)

(C) Main Boundary Thrust (MBT)

(D) Himalayan Frontal Thrust (HFT)

84. An igneous pluton which is more or less vertical and cylindrical that cross-cuts adjacent sediments and is bounded by steep faults is termed as :

(A) Laccolith

(B) Bysmalith

(C) Lopolith

(D) Conolith

85. The characteristic texture of lamporphyres is :

(A) Allotriomorphic

(B) Panidiomorphic

(C) Xenomorphic

(D) Hypidiomorphic

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86. Which of the following is **not** a monomineralic ultrabasic rock ?

(A) Anorthosite

(B) Pyroxinite

(C) Leptynite

(D) Dunite

87. The plutonic rock consisting of roughly 70% pseudoleucite (alkali feldspar, nepheline, kalsilite and minor analcime) and 30% pyroxene is termed as :

(A) Ferrilite

(B) Fenite

(C) Fergusite

(D) Felsite

88. A plutonic rock consisting of approximately equal amounts of alkali feldspar and plagioclase and with essential quartz (5%-20% of felsic minerals) but not enough to make the rock a granite is termed as :

(A) Quartz Monzonite

(B) Quartz Monzodiorite

(C) Quartz Norite

(D) Quartz Monzogabbro

89. Harzburgite is an ultramafic plutonic rock composed essentially of :

(A) Olivine + orthopyroxene

(B) Olivine + clinopyroxene

(C) Olivine + plagioclase

(D) Clinopyroxene + plagioclase

90. An intermediate volcanic rock, usually porphyritic, consisting of plagioclase (frequently zoned from labradorite to oligoclase), pyroxene, hornblende and/or biotite is termed as :

(A) Anorthosite

(B) Andesite

(C) Andesitoid

(D) Andesinite

91. Cementing material differs in different sandstones, the sandstone where cementing material is clay is named _____.

(A) Ferruginous sandstone

(B) Argillaceous sandstone

(C) Calcareous sandstone

(D) Siliceous sandstone

92. Shales are generally characterized with distinct _____.

(A) cleavage

(B) laminar structure

(C) joints

(D) parting

93. Sedimentary rocks formed due to the sediments formed from mechanical breakdown of primary rocks is called _____.

(A) siliclastic rocks

(B) clastic rocks

(C) semi-clastic rocks

(D) chemical sediments

94. A type of exfoliation due to weathering that produces rounded features when chemical weathering moves along joints in the bed is called _____.

(A) spheroidal weathering
(B) frost wedging
(C) mechanical weathering
(D) Both (A) and (C) are correct

95. Karst topography is a special type of landform characterized by sinkholes or caves, developed in _____.

(A) Calcareous sandstone
(B) Shales
(C) Carbonate bedrock
(D) Basaltic beds

96. Sedimentary rocks that are formed in arid climatic condition are classified a is _____.

(A) Ironstone
(B) Red beds
(C) Cherty sandstone
(D) All of these

97. _____ is an example of Chemical Sedimentary Rock.

(A) Breccia
(B) Laterite
(C) Bauxite
(D) Gypsum

98. The physical or chemical transformation of sediments or soil by geological processes is termed as _____.

(A) Diagenesis
(B) Petrogenesis
(C) Lithification
(D) Crustificaton

99. High SiO_2, Al_2O_3, rounded clasts and high ZTR index indicate _____ maturity of sediments.

(A) low
(B) high
(C) moderate
(D) None of these

100. An analysis that helps to determine grain size and percentage of the individual size fractions in the specific sediment sample is called _____.

(A) Granulometric analysis
(B) Gravimetric analysis
(C) Sedimentation analysis
(D) Both (A) and (B) are correct

101. Formulae and verbal scales for graphic size parameters are provided by Folk and Ward (1957). As per the relation, inclusive graphic kurtosis is calculated by which formula ?

(A) $K_G = \phi_{95} - \phi_5/2.44\ (\phi_{95} - \phi_{25})$
(B) $K_G = \phi_{95} - \phi_5/2.44\ (\phi_{75} - \phi_{25})$
(C) $K_G = \phi_{95} - \phi_5/1.44\ (\phi_{95} - \phi_{25})$
(D) Both (A) and (B) are correct

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102. Sandstone is categorized into three main types based on their variation in composition and cementing material, they include :

- Quartz Sandstone
- _____
- Litharenite or lithic sandstone

(A) Arkose
(B) Greywacke
(C) Molasse
(D) None of these

103. The origin of many heavy mineral species is limited to rather specific rock types ; hence, heavies provide information concerning _____.

(A) source rock
(B) mineral dispersion pattern
(C) quaternary sediments
(D) provenance

104. Conglomerate belongs to which type of sedimentary rocks ?

(A) Argillaceous
(B) Arenaceous
(C) Rudaceous
(D) Both (B) and (C) are correct

105. Folk (1962) described the classes of limestone in which framework grains are called allochems which include (i) _____, (ii) oolites, (iii) fossils and (iv) pellets.

(A) intraclasts
(B) micrite
(C) sparry calcite
(D) All of these

106. **Migmatite** consists of two or more constituents often layered repetitively forming a composite rock found in which metamorphic grade ?

(A) Low to medium grade
(B) Low grade
(C) Medium and high-grade
(D) High grade

107. Which will be the parental rock for **Slate** ?

(A) Shale
(B) Sandstone
(C) Limestone
(D) All of these

108. Foliation in metamorphic rocks developed due to which metamorphic process ?

(A) Thermal
(B) Dynamic
(C) Thermodynamic
(D) All of these

109. Amongst the following which is the highest grade of metamorphic rock ?

(A) Talc-schist
(B) Garnet-schist
(C) Mica-schist
(D) Chlorite-schist

110. Low grade schist is formed under which metamorphic conditions ?

(A) High temperature and local metamorphism

(B) Low temperature and regional metamorphism

(C) Low temperature and local metamorphism

(D) All of these

111. Three types of metamorphism are recognized e.g., Contact, _____, and Dynamic metamorphism. (Identify the missing type.)

(A) Thermal

(B) Static

(C) Local

(D) Regional

112. **Net-transfer reactions during metamorphic process** involve chemical components being "transferred" from one phase or set of phases to new phases produced as old ones disappear. Find out the true example of such reaction from the following :

(A) **Anorthite = grossular + kyanite + quartz**

(B) **Muscovite + quartz = K-feldspar + sillimanite + kyanite**

(C) **Biotite + quartz = orthoclase + hematite + magnetite**

(D) None of these

113. Minerals that are characteristic for a certain metamorphic grade are called _____.

(A) Typomorphic minerals

(B) Barrovian minerals

(C) Index minerals

(D) Both (A) and (C) are correct

114. The main factors that control metamorphic processes are :

(i) The mineral composition of the parent rock

(ii) The pressure and temperature at which metamorphism takes place

(iii) The types of prevailing fluids (mostly water) during metamorphism

(iv) The amount of time available for metamorphism

Find out the odd one from the listed factors :

(A) (i) and (ii) are correct

(B) (i), (ii) and (iii) are correct

(C) (ii), (iii) and (iv) are correct

(D) All of the above are true explanations

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115. The characteristic locale for blueschist type of metamorphism appears to be along a _____ being underthrust by an oceanic plate.

(A) mid-oceanic ridge

(B) continental margin

(C) transform boundary

(D) Both (A) and (C) are correct

116. Granulite facies quartzo-feldspathic plutonic rocks characterized by the presence of hypersthenes often designated as _____.

(A) Charnockite

(B) Orthoamphibolite

(C) Paramphibolite

(D) Greenalite

117. A metamorphic structure consisting of minerals like feldspar, quartz, or garnet that have been squeezed into elliptical or lens-shaped forms resembling eyes, called _____.

(A) Orthogneiss

(B) Paragneiss

(C) Augen gneiss

(D) Biotite gneiss

118. The other name of **hydrothermal metamorphism** in geological literature is _____.

(A) Autochemical metamorphism

(B) Allochemical metamorphism

(C) Metasomatism

(D) All of these

119. Common temperatures for the formation of hornfels range from about _____.

(A) 700° to $800^{\circ}C$

(B) $> 1000^{\circ}C$

(C) 1000-$1200^{\circ}C$

(D) None of these

120. Metamorphic zone developed in the contact area around igneous rock intrusions when carbonate sedimentary rocks are invaded called _____.

(A) Carbonatites

(B) Metamorphites

(C) Skarn

(D) Hornblendite

121. Which type of compressive strength is taken as the most important index property of stones ?

(A) Confined

(B) Drained

(C) Undrained

(D) Unconfined

122. Capacity of a stone to withstand bending loads is termed as :

(A) Bending strength

(B) Shear strength

(C) Transverse strength

(D) Lateral strength

123. The ratio of the total volume of the pore spaces to the total volume of the solid rock mass is termed as :

(A) Void ratio

(B) Porosity

(C) Pores ratio

(D) Effective porosity

124. The mass per unit volume of a rock sample with natural moisture content is its :

(A) Dry density

(B) Wet density

(C) Bulk density

(D) Natural density

125. Find out the incorrect statement regarding Schmidt Hammer :

(A) It is concrete test hammer originally designed for carrying out in-situ tests on the hardness of concrete.

(B) The instrument measures the distance of rebound of a controlled impact on a rock surface.

(C) The distance of rebound (R) gives an absolute measure of surface hardness or strength.

(D) Schmidt hammer rebound values have been found to correlate well with other measures of rock strength, including Young's Modulus and Uniaxial Strength.

126. The 'Brazilian test' is used to determine :

(A) Tensile strength

(B) Compressive strength

(C) Shear strength

(D) Modulus of elasticity

127. Which of the following is **not** a Natural Pozzolanic Material ?

(A) Calcined diatomaceous earths

(B) Volcanic ash, tuffs and pumicites

(C) Fly ash

(D) Opaline cherts

128. This specific grouting technique is now a days accepted worldwide as a soil improvement method which is applied for improving the strength, deformation properties and permeability of the soil ; slope stability ; liquefaction mitigation ; vibration reduction along the road, railroad and bridge foundations.

(A) Permeation Grouting

(B) Hydro Fracture Grouting

(C) Deep Mixing Methods (DMM)

(D) Compensation Grouting

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129. Which of the following In-Situ Field Testings carried out under natural in-place conditions is **not** correct ?

(A) Standard Penetration Test (SPT)

(B) Dutch Cone Penetrometer Test

(C) Direct Pull Test

(D) Water Pressure Test

130. In a resistivity survey, the current and potential electrodes are such arranged that the distances between the potential electrodes and the centre of the array are less than those between the current and potential electrodes. What type of configuration is this ?

(A) Wenner

(B) Schlumberger

(C) Dipole-dipole

(D) Pole-dipole

131. All geographic features on the Earth's surface can be characterized and defined as :

(A) Point feature

(B) Line feature

(C) Polygon feature

(D) All of these

132. As the scale of a map increases, e.g., 1 : 100000 to 1 : 50000, the relative size of the features will :

(A) Decrease

(B) Increase

(C) Remain similar

(D) No change in size

133. GIS represents a location in _____ dimensional co-ordinates.

(A) 2

(B) 3

(C) 4

(D) 5

134. The input to geoprocessing is :

(A) Alphanumeric

(B) Numeric

(C) Alphabet

(D) Datasheet

135. Which digital data creation methods are used for GIS data creation ?

(A) Digitization

(B) Orthorectified imagery

(C) Demodulation

(D) Both (A) and (B)

136. GIS accuracy depends on :

(A) Encoded processes

(B) Source data

(C) Both (A) and (B)

(D) None of these

137. Which of the following might be considered the fourth dimension in GIS ?

(A) Space

(B) Scale

(C) Time

(D) Location

138. Which of the following is not a raster data structure ?

(A) Quadtree

(B) Spaghetti

(C) Run-length encoding

(D) Block encoding

139. Spatial databases are also known as :

(A) Geodatabase

(B) Mono database

(C) Concurrent database

(D) None of these

140. SAR and LiDAR data have a resolution varying from 100-150 m.

(A) True

(B) False

(C) Both True

(D) Both False

141. The total flow in any section of a water table contour map, in which the flow nets are drawn, can be directly computed with the help of the :

(i) Geometry of the flow nets

(ii) Hydraulic conductivity

(iii) Specific yield

(iv) Head loss

Choose the correct set of answer from the following :

(A) (i), (ii), (iii)

(B) (i), (iii), (iv)

(C) (ii), (iii), (iv)

(D) (i), (ii), (iv)

142. Factors controlling the rate of Evapo-transpiration are :

(i) Soil, air and water temperature

(ii) Quantity of air and soil moisture

(iii) Wind and atmospheric pressure

(iv) Nature and state of vegetation

Choose the correct set of answer from the following :

(A) (i), (ii), (iii)

(B) (i), (ii), (iv)

(C) (i), (iii), (iv)

(D) (i), (ii), (iii), (iv)

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143. The correct arrangement of aquifers in order of their increasing storage coefficient value is :

(A) Unconfined aquifer-Semi-confined aquifer-Confined aquifer

(B) Semiconfined aquifer-Confined aquifer-Unconfined aquifer

(C) Confined aquifer-Semi-confined aquifer-Unconfined aquifer

(D) Confined aquifer-Unconfined aquifer-Semiconfined aquifer

144. Which of the following statements w.r.t. Constant Head Permeameter is **not correct** ?

(A) It is used to measure permeability of an aquifer sample in the laboratory.

(B) Undisturbed samples are collected from the field locations.

(C) It is suitable for measuring permeability of relatively fine-grained samples.

(D) The volume of water collected in a beaker after passing through the sample is measured and the time required is noted.

145. Choose the correct answer regarding the shape and the nature of the fresh water-saline water interface near to a coastal area from the options given below :

(A) Linear and horizontal

(B) Concave upward

(C) Linear and inclined

(D) Convex downward

146. Find out the term which is not related to the pumping period of a well :

(A) Drawdown

(B) Time-drawdown data

(C) Residual drawdown

(D) Distance-drawdown data

147. Conditions under which Jacob-Cooper method cannot be used for analysis of pump test data :

(i) When the pump test is of short duration

(ii) When the pump test is of long duration

(iii) When the observation well is situated far away from the main well

(iv) When the observation well is situated nearer to the main well

Choose the correct set of answer from the following :

(A) (i) and (iii)

(B) (ii) and (iii)

(C) (ii) and (iv)

(D) (i) and (iv)

148. The suitable method of drilling in the Piedmont zone (Bhabar) occurring adjacent to the Himalayan foothills is :

(A) Rotary method

(B) Cable tool method

(C) Reverse rotary method

(D) Air rotary method

149. After completion of drilling of a water well, four different aquifer zones with the following textural characteristics have been demarcated. Select the most promise zone for withdrawl of groundwater :

(A) Zone I : Effective size value 0.55 mm and uniformity coefficient 3.60

(B) Zone II : Effective size value 0.75 mm and uniformity coefficient 2.10

(C) Zone III : Effective size value 0.65 mm and uniformity coefficient 2.60

(D) Zone IV : Effective size value 0.75 mm and uniformity coefficient 3.70

150. Water with electrical conductivity value of 2500 micro siemens/cm (at 25°C) is considered suitable for drinking purpose. When expressed this drinking water standard in terms of TDS, the corresponding TDS value (in mg/ℓ) would become :

(A) 1500 mg/ℓ

(B) 1550 mg/ℓ

(C) 1600 mg/ℓ

(D) 1650 mg/ℓ

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